Facioscapulohumeral Muscular Dystrophy (FSHD): A Comprehensive Disease Characteristics Report

Disease category: Genetic (autosomal-dominant / digenic myopathy) **Suggested ontology mapping:** MONDO:0008028 (facioscapulohumeral muscular dystrophy); OMIM 158900 (FSHD1), 158901 (FSHD2); Orphanet ORPHA:269; MeSH D020391 ("Muscular Dystrophy, Facioscapulohumeral"); ICD-10 G71.0; ICD-11 8C71.

Summary

Facioscapulohumeral muscular dystrophy (FSHD) is one of the most common inherited myopathies (prevalence ~1 in 8,000–20,000) and is defined clinically by slowly progressive, characteristically **asymmetric** weakness that begins in the muscles of the face, scapular stabilizers, and upper arms and later descends to the trunk and lower limbs. Despite decades of genetic confusion, FSHD is now understood as a **single-mechanism disease**: the aberrant, epigenetically de-repressed expression of the germline transcription factor **DUX4** (double homeobox 4) in skeletal muscle, where it is normally silenced. DUX4 misexpression is reached by two genetic routes that converge on the same toxic endpoint: **FSHD1 (~95%)**, caused by contraction of the D4Z4 macrosatellite repeat array to 1–10 units on a permissive 4qA haplotype at 4q35; and **FSHD2 (~5%)**, a digenic disease requiring a pathogenic variant in a chromatin repressor (most often *SMCHD1*, also *DNMT3B*, *LRIF1*) together with a borderline-length array and a permissive haplotype [PMID: 38955828; 34711481; 37703328].

Once de-repressed, DUX4 acts as a potent transcription factor that reactivates an early-embryonic/germline program — including *ZSCAN4*, *PRAMEF* family, *TRIM43*, *MBD3L*, endogenous retrotransposons, and innate-immune mediators such as *DEFB103* — and kills muscle cells primarily through caspase-3/7-dependent apoptosis, while also inhibiting nonsense-mediated decay (NMD) via UPF1 degradation and repressing the myogenic regulators *MYOD1* and *MYF5* [PMID: 22892954; 22209328; 25564732; 30446688; 37973788]. DUX4 is

expressed in a rare, stochastic, **burst-like** fashion in a small subset of myonuclei, which explains how sporadic expression produces generalized wasting and why DUX4 itself is such a difficult pharmacodynamic biomarker [PMID: 30445587].

Clinically, FSHD has near-normal life expectancy but imposes a heavy quality-of-life burden through pain, fatigue, impaired mobility (~20% eventually require a wheelchair), sleep disturbance, and extramuscular comorbidities (high-frequency hearing loss, Coats-like retinal vasculopathy, and — in severe infantile cases — seizures and cognitive impairment) [PMID: 40879179; 38968057; 41037169]. Diagnosis is molecular, integrating D4Z4 repeat sizing, 4qA haplotyping, methylation profiling, and exome sequencing for FSHD2 [PMID: 42498520; 36348371]. **No disease-modifying therapy is approved**: the lead candidate losmapimod (a p38 MAPK inhibitor) failed both its phase 2b DUX4-expression endpoint and its subsequent phase 3 clinical endpoint, leaving multidisciplinary symptomatic care as the standard while a rich pipeline of DUX4-lowering RNA and small-molecule therapeutics advances [PMID: 38631764; 40580827; 28273791; 35884928].

This report synthesizes 13 confirmed findings across 59 reviewed papers and is organized to match the 15-section disease-characteristics template.

Key Findings

Finding 1 — Two genetic routes, one mechanism (DUX4 de-repression)

FSHD is caused by aberrant DUX4 expression reached by two mutually convergent genetic mechanisms. **FSHD1 (~95% of cases)** results from a pathogenic contraction of the D4Z4 macrosatellite repeat array to 1–10 repeat units in the subtelomeric region of chromosome 4 (4q35); normal arrays span roughly 8 to >100 units. Contraction produces chromatin relaxation and de-repression of the DUX4 retrogene that is embedded in the distal D4Z4 unit. **FSHD2 (~5%)** is a **digenic** disease: a pathogenic variant in a D4Z4 chromatin repressor — most often *SMCHD1* on chromosome 18, and also *DNMT3B* and *LRIF1* — combined with a borderline array (8–20 units) and a permissive haplotype. Both routes converge on epigenetic derepression and toxic DUX4 expression in skeletal muscle. *(Evidence: human clinical / review.)*

> *"In 95% of cases, FSHD patients carry a pathogenic contraction of the D4Z4 repeat units (RUs) in the subtelomeric region of chromosome 4 (4q35), which leads to the expression of DUX4 retrogene, toxic for muscles (FSHD1). Five percent of patients display the same clinical phenotype in association with a mutation in the SMCHD1 gene located in chromosome 18"* — PMID: 38955828
>
> *"FSHD2 is a digenic disease and that mutations in the genes SMCHD1, DNMT3B, and more recently LRIF1, can cause FSHD2"* — PMID: 34711481

Finding 2 — DUX4 is an embryonic transcription factor toxic to muscle via apoptosis and NMD inhibition

DUX4 is a double-homeobox retrogene normally expressed only in the testis and early embryo (where it drives zygotic genome activation) and epigenetically silenced in somatic tissue. When misexpressed in muscle it activates early-embryonic/germline genes and retroviral elements and drives cell death **primarily through caspase-3/7-dependent apoptosis**. DUX4 protein also proteolytically degrades UPF1, a central component of the nonsense-mediated decay (NMD) machinery; because DUX4 mRNA is itself an NMD substrate, this creates a **double-negative feedback loop** that stabilizes DUX4 and contributes to its burst-like expression. At low levels, DUX4 additionally represses the myogenic regulators *MYOD1* and *MYF5*, impairing differentiation. *(Evidence: in vitro / model.)*

> *"DUX4-triggered proteolytic degradation of UPF1, a central component of the nonsense-mediated decay (NMD) machinery, is associated with profound NMD inhibition"* — PMID: 25564732
>
> *"cell death does primarily occur through caspase 3/7-dependent apoptosis"* — PMID: 37973788
>
> *"a large set of human myogenic genes is rapidly deregulated by DUX4, including MYOD1 and MYF5"* — PMID: 30446688

Finding 3 — Extramuscular comorbidities: hearing loss, retinal vasculopathy, seizures

In the population-based MD STARnet cohort (n=548), 17.2% of patients had at least one of three key comorbidities: **hearing loss 13%** (n=71, most common), **retinal abnormalities 3.6%** (n=20), and **seizures 2.0%** (n=11). Median age at diagnosis differed markedly — hearing loss 46.5 y, retinal 58.7 y, and seizures 16.5 y (half of seizure cases occurred at ≤10 y). These extramuscular features (hearing loss, Coats-like retinal vasculopathy, cognitive impairment, spinal deformity) are enriched in severe early-onset/infantile FSHD, which carries the shortest D4Z4 repeats. *(Evidence: human clinical, population-based.)*

> *"Hearing Loss (13%; n = 71) was the most frequently reported comorbidity, followed by retinal abnormalities (3.6%; n = 20) and seizures (2.0%; n = 11)"* — PMID: 40879179
>
> *"Children with a severe, early-onset phenotype experience higher rates of extramuscular features, including hearing loss, cognitive impairment, and spinal deformities"* — PMID: 41037169

Finding 4 — Penetrance is age-dependent and incomplete; repeat size is a weak predictor

Penetrance depends on repeat size and increases into late adulthood, with roughly 17% non-penetrant and ~25% asymptomatic carriers. In a genotype–phenotype study (n=152), **familial factors accounted for 50% of the variance in disease severity**, whereas D4Z4 repeat array size explained only **~10% overall** (~30% for facial, ~15% upper limb, ~3% leg). Unaffected carriers had significantly longer arrays than symptomatic relatives (7.3 vs 6.0 units, P<0.001). Prevalence estimates range from ~1 in 20,000 to ~1 in 8,000, making FSHD one of the most common muscular dystrophies. *(Evidence: human clinical, family-based.)*

> *"Familial factors accounted for 50% of the variance in disease severity (FSHD clinical score). The explained variance by the D4Z4 repeat array size for disease severity was limited (approximately 10%)"* — PMID: 30211448
>
> *"penetrance depends on repeat size and increases until late adulthood"* — PMID: 29997197
>
> *"Facioscapulohumeral muscular dystrophy (FSHD) is a myopathy with prevalence of 1 in 20,000"* — PMID: 33303865

Finding 5 — No approved disease-modifying therapy; losmapimod failed phase 3

Losmapimod, a selective p38 α/β MAPK inhibitor that reduces DUX4 expression in vitro, was well tolerated in phase 1/2 (NCT04004000). The phase 2b **ReDUX4** trial (NCT04003974) **did not meet its primary endpoint** of reduced DUX4-driven gene expression in muscle biopsy, although some functional measures trended favorably; the subsequent phase 3 trial failed to demonstrate clear clinical benefit. Current management remains multidisciplinary and symptomatic: physical/occupational therapy, scapular fixation surgery, and management of hearing, retinal, and respiratory complications, plus pain and fatigue control. *(Evidence: human clinical trials.)*

> *"the trial failed to demonstrate a clear clinical benefit, highlighting key challenges in drug development for the disease"* — PMID: 40580827
>
> *"The primary endpoint was change from baseline to either week 16 or 36 in DUX4-driven gene expression in skeletal muscle biopsy samples"* — PMID: 38631764

Finding 6 — ACTA1-MCM/FLExDUX4 mouse recapitulates progressive DUX4-driven pathology

The tamoxifen-inducible **ACTA1-MCM;FLExDUX4/+** bitransgenic mouse expresses low chronic DUX4 in skeletal muscle and shows progressive muscular dystrophy with DUX4-pathway molecular signatures, histological damage, and functional weakness (reduced initial force with relatively preserved power/endurance). It is the most widely used academic FSHD model and is the workhorse for testing systemic antisense DUX4-silencing therapeutics, which improve molecular and histopathological readouts including in the diaphragm. *(Evidence: model organism.)*

> *"the most commonly used by academic laboratories being ACTA1-MCM/FLExDUX4"* — PMID: 39518930

Finding 7 — The permissive 4qA haplotype supplies a polyadenylation signal that stabilizes DUX4 mRNA

Pathogenic DUX4 expression requires a permissive distal **4qA haplotype** that provides a polyadenylation signal (PAS) in the pLAM region immediately telomeric to the last D4Z4 unit; this PAS stabilizes the otherwise-degraded DUX4 transcript, enabling translation of full-length DUX4. The near-identical 10q26 D4Z4 array and the non-permissive 4qB haplotype lack a functional PAS and are non-pathogenic even when contracted. FSHD1 therefore requires **three** conditions: (1) D4Z4 contraction, (2) a 4qA permissive haplotype, and (3) reduced D4Z4 CpG methylation. Downstream, DUX4 inhibits muscle differentiation, sensitizes cells to oxidative stress, and induces atrophy. *(Evidence: review / in vitro.)*

> *"A unifying pathogenic model for FSHD emerged with the recognition that the FSHD-permissive 4qA haplotype corresponds to a polyadenylation signal that stabilizes the DUX4 mRNA"* — PMID: 27816329
>
> *"Aberrant DUX4 expression triggers a deregulation cascade inhibiting muscle differentiation, sensitizing cells to oxidative stress, and inducing muscle atrophy"* — PMID: 27816329

Finding 8 — Muscle MRI reveals a specific asymmetric involvement pattern useful for diagnosis and natural history

Whole-body muscle MRI (30 FSHD vs 23 other myopathies) identified an FSHD-specific pattern of fatty replacement and atrophy, most frequently in the **trapezius, teres major, and serratus anterior**; asymmetric involvement was significantly higher in FSHD than in other myopathies. Prospective quantitative MRI shows an average yearly fat-fraction increase of **~2%** (thigh) and **~1.9%** (leg); muscles with intermediate baseline fat fraction (15–30%) or elevated water-T2 (edema >41 ms) progress fastest, and edematous muscles are at higher risk of irreversible fatty replacement. wT2 and fat fraction correlate with clinical scales, making MRI a sensitive trial biomarker. *(Evidence: human clinical imaging.)*

> *"The most frequently affected muscles, including paucisymptomatic and severely affected FSHD patients, were trapezius, teres major and serratus anterior. Moreover, asymmetric muscle involvement was significantly higher in FSHD as compared to NFSHD patients"* — PMID: 26115655

> *"The average yearly increase in FF was 2 ± 0.6% at thigh level and 1.9 ± 0.7% at leg level"* — PMID: 40172709

Finding 9 — DUX4 is expressed in rare, burst-like fashion in few myonuclei

Single-cell RNA-seq of patient-derived primary myocytes identified a small FSHD-specific cell population expressing DUX4 in a dynamic, **burst-like** manner; downstream DUX4-target activation persists long after DUX4 itself has faded. Pseudotime trajectory modeling reconstructed an FSHD cellular progression from an early DUX4 burst to downstream pathway activation (oxidative stress, immune/germline programs, apoptosis). This sporadic, stochastic expression (few DUX4-positive nuclei at any moment) dilutes the bulk-tissue signal and explains the difficulty of using DUX4 as a pharmacodynamic biomarker. *(Evidence: in vitro, single-cell.)*

> *"DUX4 has been shown to be expressed in a highly dynamic burst-like manner, likely resulting in the detection of the downstream cascade of events long after DUX4 expression itself has faded"* — PMID: 30445587

Finding 10 — DUX4 activates a germline/retrotransposon/immune program

DUX4 directly binds and activates genes of germline and early stem-cell development (*ZSCAN4*, *PRAMEF* family, *TRIM43*, *MBD3L*) plus LTR/MaLR endogenous retrotransposons; these DUX4-target genes are reliably detected in FSHD muscle but not controls, providing direct causal support for DUX4 misexpression. DUX4 also modulates innate immunity by activating *DEFB103* (β-defensin 3), which itself inhibits muscle differentiation. Loss of D4Z4 silencing manifests as CpG hypomethylation plus loss of repressive chromatin marks (H3K9me3, H3K27me3); an Argonaute-dependent siRNA pathway normally helps repress D4Z4. Notably, DUX4 is co-opted by herpesviruses and papillomaviruses, which induce it to mimic zygotic genome activation and prevent silencing of the viral genome. *(Evidence: in vitro / molecular.)*

> *"we show that DUX4 binds and activates LTR elements from a class of MaLR endogenous primate retrotransposons and suppresses the innate immune response to viral infection, at least in part through the activation of DEFB103, a human defensin that can inhibit muscle differentiation"* — PMID: 22209328

> *"DUX4 is a germline transcription factor and its expression in skeletal muscle leads to activation of early stem cell and germline programs and transcriptional activation of retroelements"* — PMID: 22892954
>
> *"Loss of D4Z4 repression in FSHD is observed as hypomethylation of the array accompanied by loss of repressive chromatin marks"* — PMID: 26113644

Finding 11 — Therapeutic pipeline targets DUX4 at multiple levels

Because no disease-modifying therapy is approved, strategies target the DUX4 axis at multiple points: (1) **transcript knockdown** with antisense oligonucleotides (AOs) directed at DUX4 mRNA/its polyadenylation signal, and RNA interference/siRNA (the Argonaute-dependent D4Z4 silencing pathway is exploitable); (2) **small molecules** suppressing DUX4 transcription (p38 MAPK inhibitors such as losmapimod — phase 3 negative; BET inhibitors; β2-agonists); (3) **blocking downstream toxicity** (flavones and other compounds reduce DUX4-induced apoptosis via an mTOR-independent mechanism); and (4) **epigenetic re-silencing**. Systemic AO therapy in the ACTA1-MCM/FLExDUX4 mouse improved molecular and histopathological readouts, including in the diaphragm. Clinical candidates include siRNA approaches (e.g., del-desiran / AOC-1020 antibody-siRNA conjugate) and other DUX4-lowering agents. *(Evidence: model organism / in vitro / early clinical.)*

> *"Antisense Oligonucleotides Used to Target the DUX4 mRNA as Therapeutic Approaches in FaciosScapuloHumeral Muscular Dystrophy"* — PMID: 28273791
>
> *"Long-Term Systemic Treatment of a Mouse Model Displaying Chronic FSHD-like Pathology with Antisense Therapeutics"* — PMID: 35884928
>
> *"we have identified a panel of five compounds that function downstream of DUX4 activity to inhibit DUX4-induced toxicity"* — PMID: 37973788

Finding 12 — Major quality-of-life burden despite near-normal life expectancy

FSHD is slowly progressive with generally normal or near-normal life expectancy, but causes substantial disability — roughly **20% of patients** eventually require a wheelchair. Pain and fatigue are among the most frequent and disabling symptoms: in cross-sectional neuromuscular studies, FSHD (with DM2) had significantly higher pain prevalence than other neuromuscular disorders and controls, and fatigue was the factor most consistently associated with reduced quality of life across muscular dystrophies. FSHD patients report worse sleep (insomnia) and lower mental-wellbeing QoL domains than controls. Pain medications are prescribed long-term for ~31–40% of adults, with impaired mobility the strongest correlate. **Infantile/early-onset FSHD (~10% of patients; onset <10 y)** is more severe with higher rates of extramuscular features. *(Evidence: human clinical.)*

> *"Patients with DM2 and FSHD had significantly higher levels of pain prevalence compared to other examined NMD subgroups and the control group"* — PMID: 38968057
>
> *"Pain medications were prescribed for 31.1%-40.2% of people 20 years and older"* — PMID: 40546227
>
> *"An estimated 10% of FSHD patients have an early onset (onset before 10 years of age) and are traditionally classified as infantile FSHD"* — PMID: 27530735

Finding 13 — Molecular diagnosis integrates repeat sizing, haplotyping, methylation, and exome sequencing

Definitive diagnosis is molecular, not by biopsy. First-line testing sizes the D4Z4 repeat and determines the permissive 4qA haplotype/structure using single-molecule methods (Southern/pulsed-field electrophoresis, molecular combing, or optical genome mapping); second-line testing (DR1 methylation profiling and whole-exome sequencing for *SMCHD1*/*DNMT3B*/*LRIF1*) is reserved for borderline, non-contracted, or atypical cases. In a tertiary cohort (135 referrals), FSHD was confirmed in 89.6%, of which FSHD1 90.1%, FSHD1+2 3.3%, and FSHD2 6.6%. Long-read sequencing (Nanopore/PacBio) now resolves repeat length and CpG methylation simultaneously at nucleotide resolution. Standard short-read WGS/WES and chromosomal microarray/karyotype alone cannot size the macrosatellite, and the near-identical 10q26 array and 4qA/4qB haplotypes must be distinguished. *(Evidence: human clinical / methods.)*

> *"First-line testing included single-molecule D4Z4 repeat sizing, haplotyping and structural analysis using molecular combing or optical genome mapping. DR1 methylation profiling and whole-exome sequencing (WES) were used as second-line tests in unresolved, borderline, non-contracted or clinically atypical cases"* — PMID: 42498520
>
> *"We applied Nanopore CRISPR/Cas9-targeted resequencing for the diagnosis of FSHD by simultaneous detection of D4Z4 repeat length and methylation status at nucleotide level"* — PMID: 36348371

Full Report by Template Section

1. Disease Information

FSHD is a slowly progressive hereditary myopathy defined by characteristically **asymmetric** weakness that begins in the facial (**facio-**), scapular (**scapulo-**), and upper-arm (**humeral**) muscles and later descends to the abdominal, foot-dorsiflexor, and pelvic-girdle muscles. It is typically classified as the second/third most common muscular dystrophy. Onset is usually between 15 and 30 years of age, but ranges from infancy to late adulthood [PMID: 41781309].

- **Key identifiers:** OMIM 158900 (FSHD1) and 158901 (FSHD2); Orphanet ORPHA:269; ICD-10 G71.0; ICD-11 8C71; MeSH D020391; MONDO:0008028.
- **Synonyms/alternative names:** facioscapulohumeral dystrophy, Landouzy–Dejerine muscular dystrophy, Landouzy–Dejerine syndrome, FSH muscular dystrophy, FSHMD.
- **Information source:** Aggregated disease-level resources (OMIM, Orphanet, MeSH) plus population registries (MD STARnet, Japanese FSHD registry, TREAT-NMD) and cohort/natural-history studies; both aggregated and patient-level data contributed to this report.

2. Etiology

Causal factors — genetic/epigenetic (Findings 1, 7, 10). The primary cause is epigenetic de-repression and toxic misexpression of DUX4 in skeletal muscle. FSHD1 (~95%): D4Z4 contraction to 1–10 units on a permissive 4qA haplotype at 4q35 [PMID: 38955828; 37703328]. FSHD2 (~5%): digenic — a chromatin-repressor variant (*SMCHD1*, *DNMT3B*, *LRIF1*) plus a borderline array and permissive haplotype [PMID: 34711481]. All routes require a permissive 4qA polyadenylation signal to stabilize DUX4 mRNA [PMID: 27816329].

Genetic risk factors. Shorter D4Z4 arrays and lower D4Z4 methylation increase risk and severity; the 4qA permissive haplotype is obligatory. *SMCHD1* acts as both a causal FSHD2 gene and a **modifier** of FSHD1 severity.

Environmental/lifestyle risk factors. No established environmental cause. Mechanical overuse and eccentric exercise are debated as local aggravators; there is no strong evidence they alter disease course. Age and (subtly) sex modify expression — several registries note earlier facial weakness and higher hearing-loss rates in females [PMID: 40203460].

Protective factors. Longer residual D4Z4 arrays and higher CpG methylation are protective; unaffected carriers have longer arrays than symptomatic relatives (7.3 vs 6.0 units, $P<0.001$) [PMID: 30211448]. No validated dietary or pharmacological protective factor exists.

Gene–environment interactions. DUX4 target genes overlap with the innate antiviral response, and herpesviruses/papillomaviruses actively induce DUX4 to promote replication [PMID: 38168299] — a biologically intriguing but not clinically established gene–environment axis.

3. Phenotypes

Phenotype	Type	Onset / severity / progression	Frequency	Suggested HPO
Facial weakness (orbicularis oculi/ oris)	Clinical sign	Early, often first; mild→moderate; progressive	Very frequent	HP:0000278 (facial weakness)
Scapular winging / shoulder-girdle weakness	Clinical sign	Adolescence/early adult; presenting complaint	Very frequent	HP:0003691 (scapular winging)
Asymmetric proximal upper-limb weakness	Clinical sign	Progressive, descending	Very frequent	HP:0003484
Foot drop / tibialis anterior weakness	Clinical sign	Later, descending	Frequent	HP:0009027
Abdominal weakness (Beevor sign)	Clinical sign	Variable	Characteristic	HP:0009063
Chronic pain	Symptom	Adult; disabling	Very frequent [PMID: 38968057]	HP:0012531
Fatigue	Symptom	Adult; disabling; QoL-dominant	Very frequent [PMID: 31307472]	HP:0012378
High-frequency sensorineural hearing loss	Lab/sign	Adult (median 46.5 y); infantile forms earlier	13% [PMID: 40879179]	HP:0008553 / HP:0000407
Retinal vasculopathy (Coats-like)	Sign	Adult (median 58.7 y)	3.6% [PMID: 40879179]	HP:0500049 / HP:0007843
Seizures	Sign	Childhood (median 16.5 y; ½ ≤10 y)	2.0% [PMID: 40879179]	HP:0001250
Respiratory muscle weakness	Sign	Advanced disease	~1/3 in registry [PMID: 40203460]	HP:0002747

Phenotype	Type	Onset / severity / progression	Frequency	Suggested HPO
Wheelchair dependence	Functional outcome	Late	~20% [Finding 12]	HP:0002540

Quality-of-life impact. Pain and fatigue drive QoL loss more than raw muscle weakness; FSHD scored lower than DMD on mental-wellbeing domains, and 25–81% of men with MD report sleep impairment [PMID: 31307472; 36137167].

4. Genetic / Molecular Information

- **Causal gene:** *DUX4* (double homeobox 4; HGNC:50800), a retrogene embedded in the distal D4Z4 unit at 4q35. The de-repression machinery involves *SMCHD1* (HGNC:29090; OMIM 614982), *DNMT3B* (HGNC:2979), and *LRIF1* (HGNC:25281) in FSHD2.
- **Pathogenic "variant" class:** FSHD1 is caused by a **structural/macrosatellite contraction** (1–10 D4Z4 units), not a point mutation — invisible to standard sequencing/microarray. FSHD2 involves loss-of-function point/indel variants in *SMCHD1* (most common), *DNMT3B*, or *LRIF1*, generally classified pathogenic/likely pathogenic per ACMG. In-cis D4Z4 duplication alleles are an emerging pathogenic class [PMID: 37703328].
- **Functional consequence:** Gain-of-toxic-function via ectopic DUX4 transcription-factor activity; the chromatin-repressor variants are loss-of-function.
- **Modifier genes:** *SMCHD1* modifies FSHD1 severity; residual repeat length and methylation modify penetrance.
- **Epigenetics:** D4Z4 CpG hypomethylation and loss of repressive H3K9me3/H3K27me3 marks are central [PMID: 26113644].
- **Chromosomal features:** 4q35 D4Z4 macrosatellite; homologous non-pathogenic 10q26 array; 4qA (permissive) vs 4qB (non-permissive) haplotypes.

5. Environmental Information

There are no established environmental, occupational, toxic, or infectious **causes** of FSHD — it is a monogenic/digenic epigenetic disease. Of mechanistic interest, DUX4 is actively induced by herpesviruses (α/β/γ) and papillomaviruses to mimic zygotic genome activation and evade viral-genome silencing, linking DUX4 biology to viral infection but not establishing infection as an FSHD trigger [PMID: 38168299]. Lifestyle factors (exercise type/intensity) are studied for symptom management, not causation.

6. Mechanism / Pathophysiology

Causal chain (upstream → downstream):

```
D4Z4 contraction (FSHD1)  ┐
   OR                     ├─► chromatin relaxation ─► DUX4 de-repression
SMCHD1/DNMT3B/LRIF1 LOF   ┘        (needs 4qA PAS to stabilize mRNA)
(FSHD2, borderline array)
        │
        ▼
  DUX4 protein (burst-like, few nuclei)
        │
   ┌────┼───────────────┬───────────────┬──────────────┐
   ▼    ▼               ▼               ▼              ▼
germline/    retro-     UPF1 degradation  MYOD1/MYF5    oxidative
stem-cell    transposon  → NMD inhibition  repression    stress +
program      activation  → DUX4 mRNA      → impaired     DEFB103
(ZSCAN4,     (LTR/MaLR)   stabilized       myogenesis    immune
PRAMEF,                    (feedback loop)                 activation
TRIM43)          │
        └────┴───► caspase-3/7 apoptosis ─► myofiber death
                                │
                                ▼
             progressive, asymmetric fatty replacement & weakness
                     (trapezius, teres major, serratus anterior first)
```

- **Molecular pathways:** p38 MAPK (drives DUX4 transcription; drug target), NMD pathway (inhibited via UPF1), apoptotic caspase cascade, innate antiviral/interferon program.
- **Cellular processes:** apoptosis (GO:0006915), impaired myogenic differentiation (GO:0007517), oxidative-stress response (GO:0006979), NMD (GO:0000184).
- **Protein dysfunction:** DUX4 gain-of-toxic-function (aberrant transcription-factor activity); SMCHD1/DNMT3B/LRIF1 loss-of-function.
- **Immune involvement:** DUX4 activates *DEFB103* (β-defensin 3), which inhibits muscle differentiation; broader innate-immune/interferon dysregulation [PMID: 22209328].
- **Molecular profiling:** DUX4-target gene panels (ZSCAN4, TRIM43, LEUTX, PRAMEF, MBD3L) are the standard transcriptomic biomarker; single-cell RNA-seq reveals rare burst-like DUX4+ cells [PMID: 30445587].
- **GO/CL suggestions:** biological process GO:0006915 (apoptotic process), GO:0007517 (muscle organ development); cell type CL:0000187 (muscle cell), CL:0000515 (skeletal muscle myoblast), CL:0000594 (skeletal muscle satellite cell).

7. Anatomical Structures Affected

- **Primary organ/tissue:** skeletal muscle (UBERON:0001134), especially facial muscles (UBERON:0001577), trapezius (UBERON:0001496), teres major, serratus anterior, biceps/triceps; later abdominal, tibialis anterior (UBERON:0001385), and pelvic girdle.
- **Secondary/systemic:** cochlea/inner ear (hearing loss; UBERON:0001846), retina (UBERON:0000966; Coats-like vasculopathy), CNS (seizures in severe infantile forms), respiratory muscles/diaphragm (UBERON:0001103).
- **Cell types:** skeletal myofibers, myoblasts, and satellite cells (CL:0000594).
- **Subcellular:** nucleus (GO:0005634; DUX4 transcription-factor site) and mitochondria/oxidative-stress machinery.
- **Lateralization:** characteristically **asymmetric/bilateral-asymmetric** — a strong diagnostic clue distinguishing FSHD from most other myopathies [PMID: 26115655].

8. Temporal Development

- **Onset:** typically 15–30 years (adolescent/young-adult); ~10% infantile/early-onset (<10 y) with more severe, extramuscular-rich disease; late-onset forms also occur [PMID: 41781309; 27530735].
- **Onset pattern:** insidious, chronic.
- **Progression:** slow, descending, often stepwise with periods of apparent stability; quantitative MRI shows ~2%/yr fat-fraction increase [PMID: 40172709].
- **Course:** chronic, lifelong, progressive; ~20% reach wheelchair dependence.
- **Critical periods:** infantile onset marks a vulnerability window; edematous (high-T2) muscles represent a window where intervention might prevent irreversible fatty replacement [PMID: 40172709].

9. Inheritance and Population

- **Epidemiology:** prevalence ~1 in 8,000–20,000 [PMID: 33303865].
- **Inheritance:** autosomal dominant (FSHD1); FSHD2 is digenic (dominant *SMCHD1* variant + permissive borderline array).
- **Penetrance:** incomplete and age-dependent (~17% non-penetrant, ~25% asymptomatic carriers) [PMID: 29997197].
- **Expressivity:** highly variable, even within families; familial factors explain ~50% of severity variance vs ~10% for repeat size [PMID: 30211448].

- **De novo / mosaicism:** a substantial minority of FSHD1 cases are de novo, frequently with somatic/germline mosaicism.
- **Sex ratio:** roughly equal, with some registries reporting earlier facial weakness and more hearing loss in females [PMID: 40203460].
- **Population diversity:** D4Z4 repeat-number distributions differ across populations (e.g., Japanese registry data) [PMID: 40203460].

10. Diagnostics (Finding 13)

- **Definitive test:** molecular — single-molecule D4Z4 repeat sizing + 4qA haplotyping (Southern/PFGE, molecular combing, or optical genome mapping); methylation profiling and WES (*SMCHD1*/*DNMT3B*/*LRIF1*) as second-line [PMID: 42498520].
- **Long-read sequencing:** Nanopore/PacBio simultaneously resolve repeat length and methylation at nucleotide resolution [PMID: 36348371].
- **Supportive tests:** muscle MRI (asymmetric trapezius/teres major/serratus anterior fatty replacement; wT2 edema) [PMID: 26115655; 40172709]; serum CK mildly elevated or normal; EMG myopathic; biopsy non-specific and not required.
- **Biomarkers:** DUX4-target gene panel in muscle (research/trial pharmacodynamic marker); MRI fat fraction and water-T2 as imaging biomarkers.
- **Differential diagnosis:** limb-girdle muscular dystrophies, Pompe disease, mitochondrial myopathy, scapuloperoneal syndromes, polymyositis — distinguished by the asymmetric facioscapular pattern and molecular testing.
- **Screening:** cascade genetic testing of at-risk relatives; prenatal/preimplantation diagnosis feasible where a defined pathogenic allele exists.

11. Outcome / Prognosis

- **Life expectancy:** generally normal/near-normal.
- **Morbidity:** substantial; ~20% wheelchair-dependent; respiratory insufficiency in advanced/infantile disease (~1/3 with respiratory dysfunction, some ventilator-dependent in registry) [PMID: 40203460].
- **QoL:** dominated by pain and fatigue; worse mental-wellbeing and sleep than controls [PMID: 38968057; 31307472; 36137167].
- **Prognostic factors:** shorter D4Z4 repeats, lower methylation, infantile onset, and *SMCHD1* co-variants predict greater severity; MRI water-T2/fat fraction predict local progression [PMID: 40172709].

12. Treatment (Findings 5, 11)

- **No approved disease-modifying therapy.** Losmapimod (p38 MAPK inhibitor) failed phase 2b (DUX4-expression endpoint) and phase 3 (clinical endpoint) [PMID: 38631764; 40580827].
- **Standard of care (symptomatic/supportive; NCIT: supportive care, physical therapy, occupational therapy):** physical/occupational therapy, aids/orthotics (ankle-foot orthoses for foot drop), scapular fixation/arthrodesis surgery, respiratory support, aggressive pain and fatigue management, and screening/treatment of hearing and retinal complications.
- **Pipeline (DUX4-axis):** antisense oligonucleotides against DUX4 mRNA/PAS [PMID: 28273791]; siRNA (e.g., del-desiran/AOC-1020 antibody-siRNA conjugate); small molecules suppressing DUX4 transcription (BET inhibitors, β2-agonists); downstream-toxicity blockers (flavones, mTOR-independent) [PMID: 37973788]; epigenetic re-silencing. Systemic AO improved molecular/histopathology (incl. diaphragm) in the ACTA1-MCM/FLExDUX4 mouse [PMID: 35884928].
- **Pharmacogenomics:** none established.

13. Prevention

- **Primary prevention:** not possible (genetic disease); genetic counseling and reproductive options (prenatal/PGT) for at-risk families.
- **Secondary prevention:** cascade genetic testing; audiology and dilated retinal exams for early detection of hearing loss and Coats-like vasculopathy, especially in short-repeat/ infantile cases.
- **Tertiary prevention:** physical therapy, respiratory surveillance, fall prevention, pain/ fatigue management, and orthopedic management of scapular winging and spinal deformity.
- **Counseling:** genetic counseling essential given incomplete penetrance, variable expressivity, de novo/mosaic cases, and digenic FSHD2 inheritance.

14. Other Species / Natural Disease

FSHD is essentially a **human-specific disease** at the genetic level: the DUX4 retrogene and the 4q35 D4Z4 macrosatellite architecture (with the permissive 4qA polyadenylation signal) are primate/human features, so no naturally occurring animal FSHD equivalent is recognized. DUX4 has orthologs/paralogs (the *Dux* family) in mouse and other mammals involved in zygotic genome activation, enabling comparative study of DUX4 biology, but not a natural disease model. No established zoonotic or veterinary relevance. NCBI Taxon: *Homo sapiens* (9606).

15. Model Organisms (Finding 6)

- **Principal model:** the tamoxifen-inducible **ACTA1-MCM;FLExDUX4**/+ bitransgenic mouse — low chronic muscle DUX4, progressive dystrophy, DUX4-pathway molecular signature, histological damage, and functional weakness; the academic workhorse for testing DUX4-silencing AOs [PMID: 39518930; 35884928].
- **Other systems:** DUX4-inducible and AAV-DUX4 mouse models; patient-derived primary myoblasts/myotubes and iPSC-derived muscle (used for single-cell RNA-seq and drug screening) [PMID: 30445587; 37973788]; zebrafish DUX4-toxicity models.
- **Model type:** tunable overexpression models — strength is faithful DUX4-pathway recapitulation and diaphragm involvement; limitation is that they model DUX4 toxicity rather than the native human D4Z4 macrosatellite contraction/epigenetic context, and burst-like stochastic expression is hard to reproduce exactly.
- **Applications:** preclinical testing of DUX4-lowering therapeutics, biomarker development, and mechanism dissection.

Mechanistic Model / Interpretation

The unifying model of FSHD is elegant: **all genetic routes lead to the same molecular villain, DUX4.** FSHD1 and FSHD2 differ only in *how* the D4Z4 array loses its repressive chromatin state (cis-contraction vs trans-acting repressor loss), but both require a permissive 4qA haplotype whose polyadenylation signal rescues the otherwise-unstable DUX4 mRNA (Finding 7). This explains a long-standing paradox — why an identical, near-homologous D4Z4 array on chromosome 10q26 is harmless, and why 4qB alleles are non-pathogenic.

Once expressed, DUX4 behaves as the germline master-regulator it evolved to be, switching on an early-embryonic/germline/retrotransposon program in a tissue where that program is catastrophic (Findings 2, 10). Muscle nuclei die by caspase-3/7 apoptosis; a self-reinforcing UPF1/NMD feedback loop stabilizes DUX4 into "bursts"; and *MYOD1*/*MYF5* repression cripples regeneration. Crucially, DUX4 is expressed in only a **tiny, stochastic subset of nuclei at any instant** (Finding 9), yet its downstream damage persists and accumulates — reconciling how a rarely-detectable transcript produces relentless, generalized wasting, and why DUX4 itself is a treacherous pharmacodynamic endpoint (a likely contributor to the losmapimod phase 2b/3 failures, Finding 5).

The clinical phenotype maps onto this biology: asymmetric involvement of trapezius/teres major/serratus anterior (Finding 8), incomplete age-dependent penetrance driven far more by familial modifiers than by repeat size (Finding 4), and a heavy pain/fatigue/mobility burden despite near-normal survival (Finding 12). The therapeutic logic follows directly: silence DUX4 (ASO/siRNA), re-repress the locus (epigenetic), or blunt downstream toxicity (Finding 11).

Evidence Base

PMID	Role in report	Supports finding(s)
38955828	French national protocol; FSHD1/FSHD2 proportions & mechanism	F1
34711481	FSHD2 digenic; SMCHD1/DNMT3B/LRIF1	F1
37703328	D4Z4 biology; in-cis duplication alleles	F1
25564732	UPF1/NMD feedback loop	F2
37973788	Caspase-3/7 apoptosis; downstream-blocking flavones	F2, F11
30446688	DUX4 represses MYOD1/MYF5	F2
40879179	MD STARnet comorbidity frequencies	F3
41037169	Extramuscular features in severe early-onset	F3
30211448	Familial vs repeat-size variance	F4
29997197	Age-dependent penetrance	F4
33303865	Prevalence 1/20,000; borderline-allele guidance	F4
38631764	ReDUX4 phase 2b design/endpoint	F5
40580827	Phase 3 losmapimod failure	F5
39518930	ACTA1-MCM/FLExDUX4 model	F6
27816329	4qA PAS stabilizes DUX4 mRNA	F7
26115655	MRI asymmetric pattern	F8
40172709	Quantitative MRI progression	F8
30445587	Single-cell burst-like DUX4	F9
22209328	Retrotransposon/DEFB103 immune activation	F10
22892954	Germline/stem-cell program	F10
26113644	D4Z4 hypomethylation / chromatin	F10
28273791	ASO targeting DUX4 mRNA	F11

PMID	Role in report	Supports finding(s)
35884928	Systemic AO efficacy in mouse	F11
38968057	Pain burden	F12
40546227	Chronic pain-medication use	F12
27530735	Infantile FSHD ~10%	F12
42498520	Two-tier diagnostic workflow	F13
36348371	Long-read repeat+methylation diagnosis	F13
38168299	Viral induction of DUX4 (GxE context)	§5
40203460	Japanese registry; sex/population differences	§3, §9, §11
31307472; 36137167	QoL, fatigue, sleep	§3, §11
41781309	Clinical overview & diagnostic pathway	§1, §8

Limitations and Knowledge Gaps

1. **DUX4 biomarker problem.** Burst-like, rare expression makes DUX4/DUX4-target quantification noisy; this likely undermined the losmapimod DUX4-expression endpoint and complicates all future trials (Findings 5, 9).
2. **Genotype–phenotype gap.** Repeat size explains only ~10% of severity; the "familial factors" that explain ~50% are largely unidentified beyond *SMCHD1* and methylation (Finding 4).
3. **Modifier biology.** The full set of severity modifiers, and the mechanisms of the striking within-family variability and asymmetry, remain unresolved.
4. **Diagnostic access.** Gold-standard sizing/haplotyping requires specialized single-molecule or long-read methods unavailable in many labs; standard WGS/WES/microarray cannot diagnose FSHD1 (Finding 13).
5. **Model fidelity.** Overexpression mouse models capture DUX4 toxicity but not the native macrosatellite/epigenetic context or the exact stochastic burst pattern (Finding 6).

6. **Therapeutic translation.** No disease-modifying therapy has yet succeeded clinically; downstream vs upstream targeting trade-offs are unsettled.
7. **Extramuscular pathophysiology.** Mechanisms of hearing loss, retinal vasculopathy, and CNS involvement (and why they cluster in short-repeat/infantile disease) are poorly defined.

Proposed Follow-up Experiments / Actions

1. **Develop robust, minimally-invasive DUX4-activity biomarkers** — e.g., blood-based or MRI-linked composite DUX4-target signatures — to serve as trial pharmacodynamic endpoints that overcome burst-like expression (addresses Findings 5, 9).
2. **Modifier-gene discovery** — large family-based WGS + methylation + long-read studies to identify the ~50% "familial" severity variance beyond *SMCHD1* (addresses Finding 4).
3. **Head-to-head preclinical benchmarking** of DUX4-lowering modalities (ASO vs siRNA/AOC vs epigenetic re-silencing vs downstream blockers) in ACTA1-MCM/FLExDUX4 mice with standardized diaphragm and functional readouts (addresses Findings 6, 11).
4. **MRI-guided trial enrichment** — use baseline water-T2/intermediate fat fraction to select fast-progressing muscles/patients and shorten trials (addresses Finding 8).
5. **Prospective natural-history + registry expansion** (TREAT-NMD/MD STARnet/Japanese registry) to refine penetrance, sex differences, and comorbidity incidence, especially for FSHD2 and infantile FSHD (addresses Findings 3, 4, 12).
6. **Broaden diagnostic access** by validating optical genome mapping and long-read sequencing as first-line clinical assays that jointly resolve repeat length, haplotype, and methylation (addresses Finding 13).
7. **Mechanistic study of extramuscular DUX4 effects** in cochlear and retinal tissue/models to explain the short-repeat comorbidity cluster.

Report compiled from 13 confirmed findings and 59 reviewed papers across 5 investigation iterations. Evidence sources span human clinical cohorts/registries, model-organism (ACTA1-MCM/FLExDUX4 mouse), in vitro (patient myoblasts, single-cell RNA-seq), and computational/review analyses.